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ACADEMY

Video 6 · Probability & Statistics Worked Solutions

Cambridge IGCSE Mathematics 0580, Paper 2 Non-calculator (Extended). Full worked solutions to Questions 24–26 of the NurNabi Academy Full Revision Paper, solved the simplest way, with the method and accuracy marks shown at every step, the way a Cambridge examiner awards them.

3	14	Q24–Q26	14:50
questions	marks	E8.3 – E9.7	video runtime

M1 method mark A1 accuracy mark (needs the M mark) B1 independent mark

Question 24

Combined events — without replacement

4 red + 6 green = 10 counters. Two taken, **without replacement**.

(a) Find the probability that both counters are green. [2]

1st green: $\frac{6}{10}$. Now 9 left, 5 of them green: $\frac{5}{9}$

$P(G, G) = \frac{6}{10} \times \frac{5}{9} = \frac{30}{90}$ M1

$\frac{1}{3}$ A1

Without replacement: the second fraction has **one fewer on top and bottom**.

(b) Find the probability that the two counters are different colours. [2]

Two ways: red then green, **or** green then red.

$P(R, G) = \frac{4}{10} \times \frac{6}{9} = \frac{24}{90}$

$P(G, R) = \frac{6}{10} \times \frac{4}{9} = \frac{24}{90}$

Add: $\frac{24}{90} + \frac{24}{90} = \frac{48}{90}$ M1

$\frac{8}{15}$ A1

Common mistake: only counting red-then-green. “Different colours” has **two** orders.

Question 25

Grouped data — estimated mean and histogram

(a) Calculate an estimate of the mean waiting time. [3]

Use the **midpoint** of each class, then $f \times$ midpoint: M1

Time	f	mid	f × mid
0 – 10	8	5	40
10 – 20	12	15	180
20 – 40	16	30	480
40 – 60	4	50	200
	40		900

Mean = $\Sigma fx \div \Sigma f = 900 \div 40$ M1

22.5 minutes A1

Why "estimate"? We don't know the exact times, so we assume everyone in a class is at the midpoint.

Watch out: the classes are **different widths**. The midpoint of 20–40 is 30, and of 40–60 is 50.

Common mistakes: dividing by 4 (the number of classes) instead of 40, or using the class width instead of the midpoint.

Non-calculator: $900 \div 40 = 90 \div 4 = 22.5$

Question 25 (continued)

Grouped data — estimated mean and histogram

(b) Complete the histogram. [3]

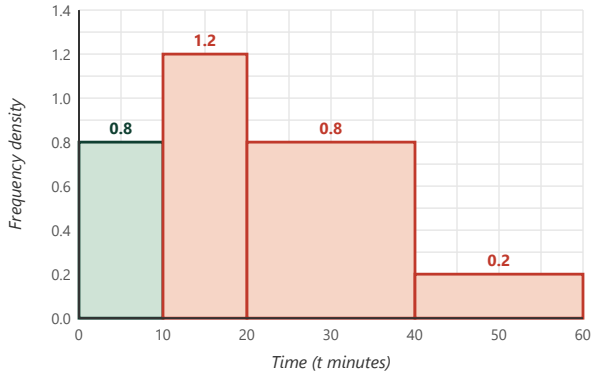
Frequency density = $\frac{\text{frequency}}{\text{class width}}$

Class	f	width	FD	
0 – 10	8	10	0.8	(given)
10 – 20	12	10	1.2	
20 – 40	16	20	0.8	
40 – 60	4	20	0.2	

Draw each bar across its **full class width**, up to its FD. **B1**

B1 for each correct bar (3 bars to draw).

Common mistake: plotting the frequency (12, 16, 4) instead of the frequency density. In a histogram, the **area** shows the frequency, not the height.



Check the given bar: 8 ÷ 10 = 0.8 ✓, which confirms you've read the scale correctly.

Question 26

Cumulative frequency

80 students. Read **across** from the frequency axis, then **down** to time.

(a) Estimate the median. [1]

- Median at $\frac{80}{2}$ = 40th student → across from 40, down to **30**

30 min B1

(b) Estimate the interquartile range. [2]

- LQ at $\frac{80}{4}$ = 20 → 20 min UQ at $\frac{3 \times 80}{4}$ = 60 → 40 min B1

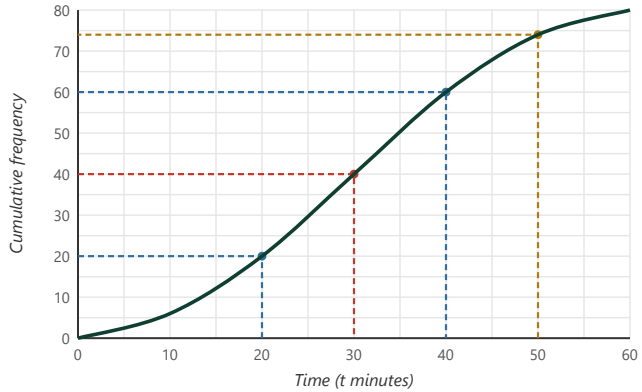
IQR = UQ − LQ = 40 − 20

20 min B1

(c) Estimate how many students spent more than 50 minutes. [1]

- Up from 50 → 74 students spent **up to** 50 min. "More than" = the rest:

80 − 74 = 6 B1



Common mistakes: reading down from 40 on the *time* axis. In (c), giving 74, which is the number who spent **less** than 50 min.

Graph readings get a small tolerance (e.g. ±1 min), so draw your lines neatly with a ruler.

Answers at a glance

Video 6 · Probability & Statistics · 14 marks

Q	Topic	Answers	Marks
24	Without replacement	(a) $\frac{1}{3}$ (b) $\frac{8}{15}$	4
25	Estimated mean, histogram	(a) 22.5 min (b) bars at FD 1.2, 0.8, 0.2	6
26	Cumulative frequency	(a) 30 min (b) 20 min (c) 6	4
	Total		**14**

Mark allocations follow the style of the Cambridge 0580 mark scheme. The questions are original NurNabi Academy practice questions, not a prediction of the real paper. That completes the full paper: **100 marks across six videos**.